

A PROJECT TITLE  
ON  
Product Demand Forecasting Using Ridge and Lasso Regression  
project report submitted in partial fulfilment of the requirements for the Course

## MACHINE LEARNING

## BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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**Koneru Lakshmaiah Education Foundation**

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF \_\_\_\_\_ CSE \_\_\_\_\_

Name of Title: Product Demand Forecasting Using Ridge and Lasso Regression

## 1. Abstract:

Product Demand Forecasting is an important application of Machine Learning that helps businesses predict the future demand for their products based on historical sales data. This project focuses on developing a demand forecasting system using Ridge Regression and Lasso Regression, which are regularized linear regression techniques. The system uses factors such as previous sales, product price, discounts, advertising expenditure, seasonal trends, and other relevant features to predict future product demand.

The dataset is first cleaned and preprocessed by handling missing values, selecting important features, and scaling numerical data. Ridge and Lasso Regression models are then trained using historical product sales information. Ridge Regression helps reduce the effect of highly correlated features, while Lasso Regression can identify and remove less important features by reducing their coefficients to zero. The performance of both models is evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and  $R^2$  Score.

The main objective of this project is to compare Ridge and Lasso Regression and determine which model provides better demand predictions. The proposed system can help businesses make better decisions regarding inventory management, production planning, purchasing, and sales strategies by providing reliable estimates of future product demand.

## 2. INTRODUCTION:

Product demand forecasting is the process of predicting the future demand for products using historical sales and other influencing factors. Accurate demand prediction is important for businesses because it helps them maintain optimal inventory levels, reduce unnecessary costs, improve production planning, and avoid stock shortages or excess inventory. Traditional forecasting methods may not effectively capture the relationships between different factors such as sales history, price, discounts, advertising expenditure, and seasonal trends. Machine Learning provides an effective approach by learning patterns from historical data and using them to make future predictions.

This project focuses on applying **Ridge Regression and Lasso Regression** for product demand forecasting. These regularized regression techniques help improve prediction accuracy and handle problems such as multicollinearity and irrelevant features. The project compares the performance of both models using standard evaluation metrics to identify the more suitable approach for predicting product demand.

## 3. SOFTWARE REQUIREMENTS:

| <b>Software / Tool</b>  | <b>Purpose</b>                                                                 |
|-------------------------|--------------------------------------------------------------------------------|
| <b>Python 3.x</b>       | Main programming language                                                      |
| <b>Jupyter Notebook</b> | Writing and executing Python code                                              |
| <b>Pandas</b>           | Data loading, cleaning, and preprocessing                                      |
| <b>NumPy</b>            | Numerical and mathematical operations                                          |
| <b>Scikit-learn</b>     | Implementing Ridge Regression, Lasso Regression, preprocessing, and evaluation |
| <b>Matplotlib</b>       | Creating graphs and visualizations                                             |
| <b>Seaborn</b>          | Statistical data visualization                                                 |
| <b>CSV Dataset</b>      | Storing historical product sales data                                          |
| <b>Web Browser</b>      | Accessing Jupyter Notebook / development environment                           |
| <b>Operating System</b> | Windows / macOS / Linux                                                        |

Signature of the Guide

Course Instructor